STUDENT TEAM WEEKLY REPORT: MODULE 4

PROJECT / FEATURE	TEAM & MENTOR	WEEK / LEADER
Question Bank & Quiz Engine [Client-Server Module, Secure Session Communication & Session Timing]	**TEAM ID:** 3 **FACULTY MENTOR:** Mr. Rudragouda Patil	**WEEK ENDING:** 31/08/2026 **STUDENT LEADER:** Mrunal Kolhe

WORK COMPLETED	CHALLENGES / RISKS	HELP NEEDED
• **Client-Server Module Implementation:** Configured Java HTTP REST server (AppServer.java) running on port 8080 using com.sun.net.httpserver.HttpServer with cached thread pool Executors.newCachedThreadPool(). • **Security for Session Comms:** Engineered secure Bearer session tokens (SessionManager.java) returning unique tokens (mmcoe_tok_<UUID>) on successful authentication. • **Session Time Recording per Client:** Implemented tracking of client session start time, duration, and active connections recorded and displayed on the admin dashboard. • **Secure HTTP Response Headers:** Injected enterprise security headers on all server responses: Strict-Transport-Security (HSTS), X-Content-Type-Options: nosniff, X-Frame-Options: DENY, and X-XSS-Protection. • **Unified API Client Bridge:** Built frontend/js/apiClient.js with automated request retries, heartbeat monitoring (/api/health), and Bearer header injection.	• **CORS Policy Blocking:** Browser blocked asynchronous fetch calls between client dashboards and port 8080. Action: Implemented global enableCORS() setting allow-origin, allow-methods, and allow-headers. **GREEN (RESOLVED)** • **Session State Inconsistency:** Server restarts wiped client session state. Action: Integrated dual-layer state syncing using memory registry with fallback to client storage. **GREEN (RESOLVED)** • **Stale Session Leaks:** Abandoned sessions consumed server resources. Action: Built rolling 2-hour timeout mechanism with periodic expiration cleanup. **GREEN (RESOLVED)**	**Client-Server Contract Verification:** Request the Integration team to test apiClient.js under simulated network drops to verify whether the 2-stage retry engine recovers seamlessly.

INTEGRATION AND COLLABORATION
Database Team: Added session tracking attributes and active user connection metrics in PostgreSQL schema.
Frontend Team: Provided unified apiClient.js and session header structure (Authorization: Bearer <token>).
Integration Team: Verified client-server handshake, heartbeat monitoring (/api/health), and REST routes via test_api_endpoints.py.
Pipeline Learnings: Gained hands-on knowledge in embedded HTTP server development, header hardening, and REST client-server protocols.

COMMON KPI EVIDENCES
Server Health & Availability: 100% uptime on /api/health heartbeat with latency <5 ms.
Security Headers Compliance: A+ rating on security response headers (HSTS, X-Frame-Options, X-Content-Type-Options).
Continuous Delivery: Code merged and validated on branch backend_client_server_mrunal.

NEXT WEEK COMMITMENTS	MENTOR RESPONSE
• **Mrunal Kolhe:** Build login attempt rate-limiter and 15-minute TIMEOUT lockout engine. (Due: Sep 03, 2026) • **Mrunal Kolhe:** Implement SubjectConcurrencyManager using ReentrantLock for simultaneous question bank uploads. (Due: Sep 05, 2026)	